Neurobehavioral Dynamics of Numerical Change

Directional and Numerical Distance Processing

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Key Findings

- Active numerosity monitoring recruits a frontal **MMN** (100–120 ms) absent in passive paradigms—indexing automatic change detection.
- **Decreasing** changes yield higher accuracy and **stronger, later MMN** than Increasing changes.
- Neural signals (MMN, N1) **scale with numerical distance**, mirroring behavioral RT and accuracy.
- **Frontal/posterior dissociation:** Decreasing $>$ Increasing for frontal MMN; Increasing $>$ Decreasing for posterior N1.

Theoretical Overview

- **Dual Systems of Number:** Parallel Individuation (PI) supports small sets (1–3); the Approximate Number System (ANS) handles larger sets (4+).
- **The Gap:** Hyde & Spelke (2012) demonstrated N1 scaling in passive tasks. Neural mechanisms of *active* numerical change detection—especially direction and distance—remain underexplored.
- **Current Study:** An active "oddball" paradigm enables analysis of frontal MMN (automatic change detection) alongside posterior N1 (sensory encoding) and P3b (working memory updating).

Experimental Design **Paradigm: Active Change Detection**

Participants monitored streams of dot arrays and pressed a key upon detecting a numerosity change. Feedback and rewards maintained engagement and vigilance.

- **Participants & Stimuli:** $N = 24$ right-handed adults; arrays of 1–6 dots (cumulative area/luminance controlled).
- **Timing:** 250 ms stimulus duration; 750–1250 ms jittered ISI.
- **Habituation:** 3–5 repeats of a prime numerosity.
- **Target:** *No Change* ($3 \rightarrow 3$), *Within-Range* (Small or Large sets), or *Crossover* ($3 \rightarrow 4$, crossing the small/large boundary).

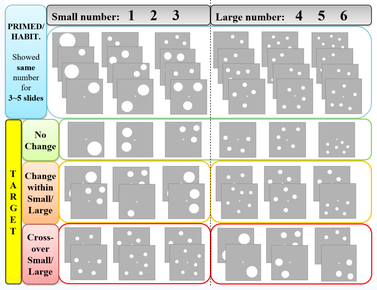


Fig. 1. Active change-detection task and trial structure.

EEG Acquisition and Processing **Recording & Preprocessing**

- 128-channel EGI Geodesic Sensor Net (250 Hz sampling rate).
- Bandpass filtered 0.1–100 Hz (online); low-pass 40 Hz (offline); re-referenced to average.
- Epochs segmented -100 to 600 ms relative to target onset.

Analysis Windows & Montages

- **P1 (80–120 ms) & N1 (125–200 ms):** Bilateral POT junction. Left: 66, 65, 59, 60, 67, 71, 70; Right: 84, 76, 77, 85, 91, 90, 83.
- **P3b (435–535 ms):** Midline Parietal (Pz). Electrodes: 62, 78, 77, 72, 67, 61, 54, 55, 79.
- **MMN (100–120 ms):** Midline Frontal (Fz); indexes automatic numerical change detection.

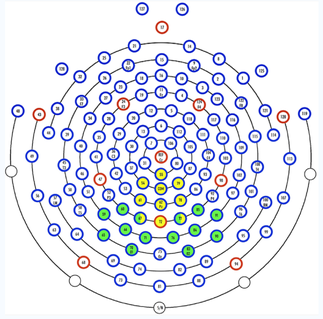


Fig. 2. Electrode map used for POT and midline analyses.

Statistical Analyses **Behavioral Data**

- Log-transformed RT modeled with **Generalized Estimating Equations (GEE)** using an exchangeable working covariance, accounting for within-subject dependence.
- Accuracy modeled via **logistic GEE**; odds ratios reported with 95% confidence intervals.
- Subject-level effects confirmed with paired t-tests on individual slopes and mean differences.

ERP Data

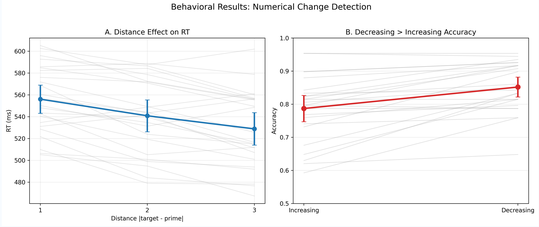
- Mean amplitude extracted within each component window (P1, N1, MMN, P3b) per participant per condition.
- Condition contrasts tested with **linear mixed models**; Bonferroni correction applied for multiple comparisons.
- Epoch acceptance threshold: $\pm 100\,\mu V$; artifact epochs excluded prior to averaging.

Component	Window	Region	Reflects
P1	80–120 ms	Bilateral POT	Early visual encoding
N1	125–200 ms	Bilateral POT	Sensory discrimination
MMN	100–120 ms	Frontal (Fz)	Auto. change detection
P3b	435–535 ms	Parietal (Pz)	WM updating

Behavioral Results: Distance & Direction

Participants monitored and reported numerical changes.

- **Distance effect:** RT decreased as numerical distance increased (Dist 3 faster than Dist 1), consistent with the standard distance effect.
- **Directional asymmetry:** Accuracy was significantly higher for Decreasing vs. Increasing changes.
- **Data included:** 24 participants, 5184 change trials for accuracy, and 4247 correct change trials with valid RT.



- **RT distance effect:** log-RT GEE slope $b = -0.0261$, $SE = 0.0033$, $p = 5.31 \times 10^{-15}$ (~2.57% faster per +1 distance).
- **Accuracy direction effect:** OR(Dec vs. Inc) $= 1.596$, 95% CI [1.339, 1.904], $p = 1.96 \times 10^{-7}$.
- **Subject-level (two-sided):** RT slope $= -13.60$ ms/step ($p = 3.05 \times 10^{-7}$); accuracy D−I $= 0.0648$ ($p = 7.67 \times 10^{-5}$).

Frontal Mismatch Negativity (MMN)

- Active monitoring recruited robust frontal MMN at Fz, peaking at $\sim$100–120 ms—indexing automatic numerical change detection.
- **Directional asymmetry:** Decreasing changes (red) elicited stronger, later MMN than Increasing (green), suggesting higher neural cost for quantity reduction.
- **Distance effect:** MMN amplitude scaled with numerical distance (Dist 3 $>$ Dist 1), mirroring behavioral accuracy.

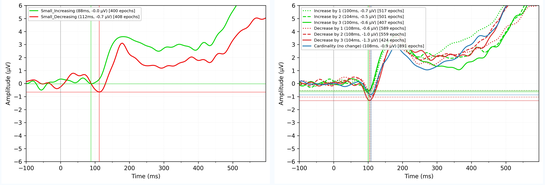


Fig. 3. Fz no-topo overlays for direction and distance effects.

Posterior N1: Direction Asymmetry

- Posterior N1 (bilateral POT) was **larger for Increasing** than Decreasing changes—opposite to frontal MMN—revealing a frontal/posterior dissociation in numerical change processing.
- P1–N1 peak-to-peak amplitude grew with numerical distance in both directions, paralleling MMN distance scaling.

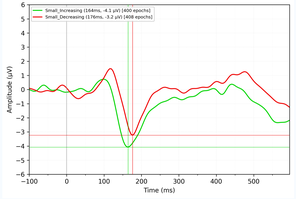


Fig. 4. N1 no-topo: Increasing vs. Decreasing direction effect (bilateral POT).

N1 Cardinality Scaling

- Replicating prior passive findings, posterior N1 over POT scaled with small-number cardinality ($1 < 2 < 3$).
- Increasing changes elicited larger N1 amplitudes than Decreasing changes.
- N1 and P1–N1 peak-to-peak patterns grew with numerical distance in both directions.

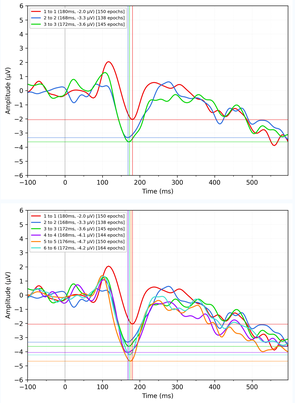


Fig. 5. N1 no-topo overlays for cardinality scaling.

P3b: Working Memory Updating

- P3b peaked $\sim$460–480 ms over parietal (Pz); **Decreasing elicited larger P3b** than Increasing—mirroring the MMN direction asymmetry.
- P3b was markedly larger for change vs. no-change ($\sim$3.1 vs. 0.5 μV), confirming working memory engagement upon numerosity change.

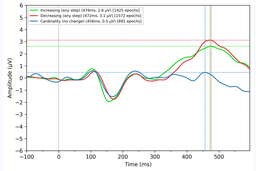


Fig. 6. P3b no-topo: direction effect and change vs. no-change (parietal Pz).

Conclusions

- **Active vs. Passive:** Active numerosity monitoring recruits rapid frontal mechanisms (MMN) distinct from posterior sensory processing (N1).
- **Direction Matters:** The brain processes increases and decreases of quantity differently.
- **Decreasing advantage:** Decreasing changes elicit higher accuracy and stronger frontal error signals (MMN).
- **Distance Effect:** Neural error signals scale with the magnitude of numerical change, mirroring behavioral performance.
- **Frontal/Posterior Dissociation:** Increasing changes drive larger posterior N1; Decreasing changes drive stronger frontal MMN—revealing dual neural processing streams for numerical change.

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Acknowledgments

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